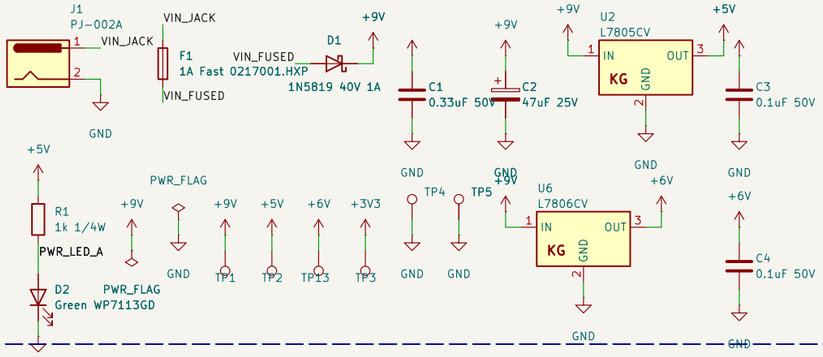
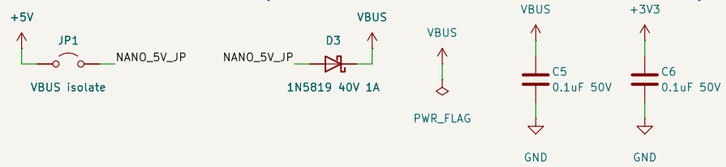
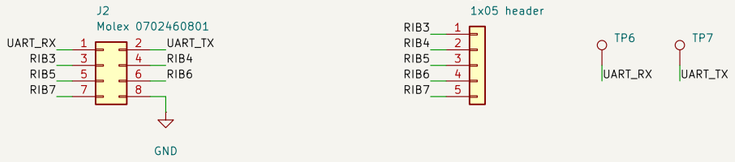
POWER: 9–12 V IN, FUSE, REVERSE PROTECTION, L7805 (5 V), L7806 (6 V MOTOR)



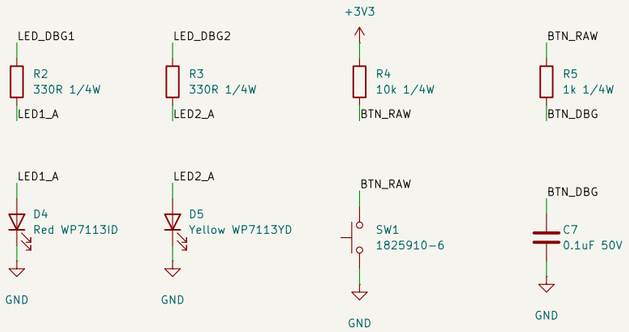
MCU POWER: 5 V TO NANO VBUS (USB BACK-FEED PROTECTION + ISOLATION JUMPER)



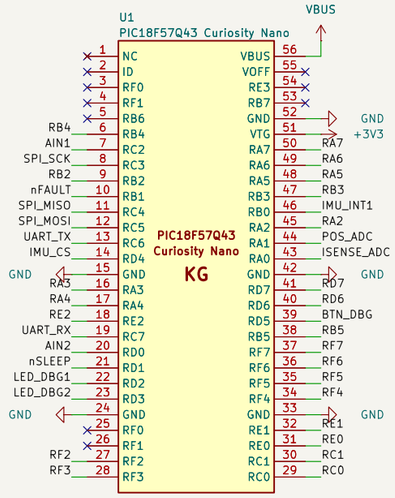
TEAM RIBBON: 2x4 IDC TO HUB CONNECTOR 3 (PIN 8 = GND)



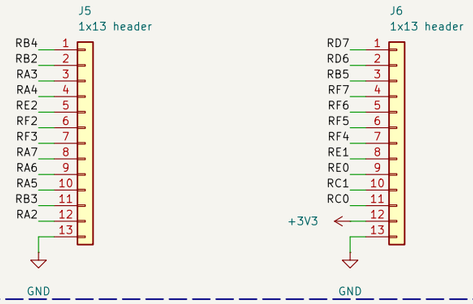
DEBUG: LEDS, HARDWARE-DEBOUNCED BUTTON



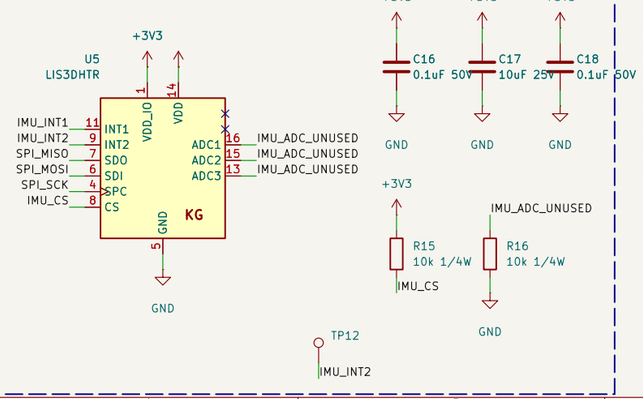
MICROCONTROLLER: PIC18F57Q43 CURIOSITY NANO



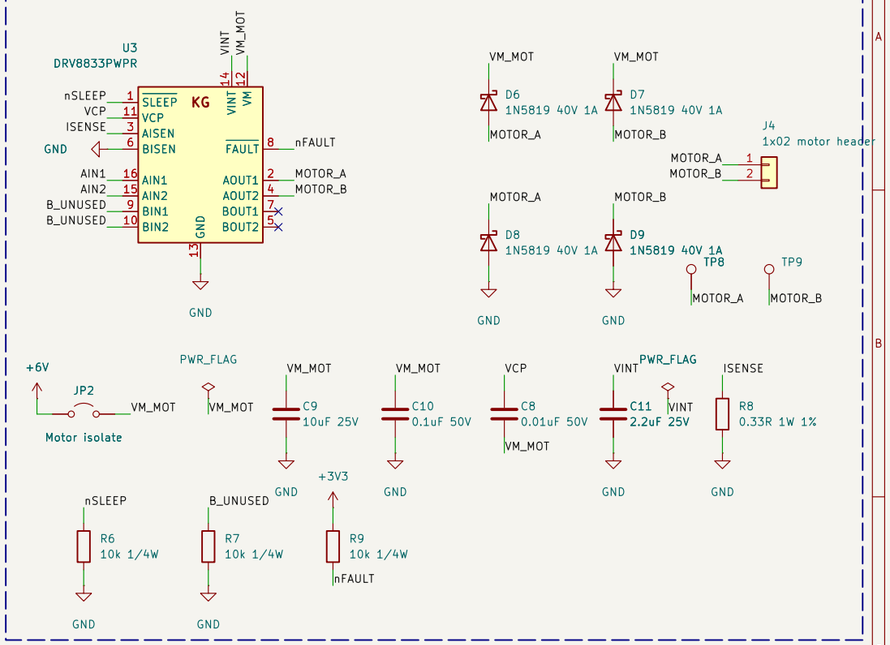
SPARE GPIO HEADERS (DEBUG / EXPANSION)



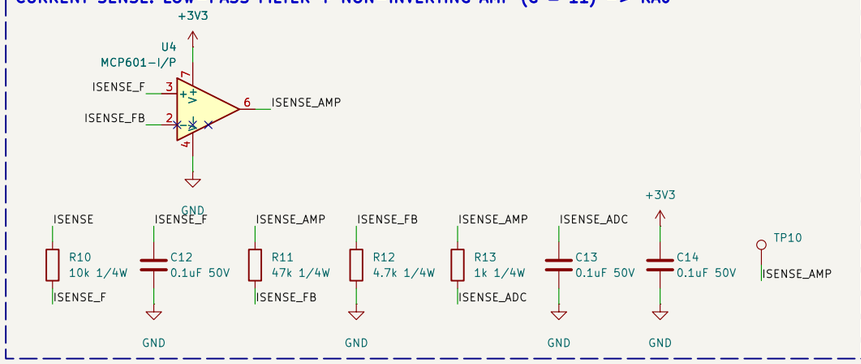
IMU: LIS3DHTR 3-AXIS ACCELEROMETER (SPI1) -> JERK / IMPULSE



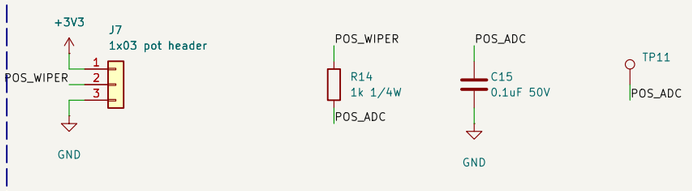
ACTUATOR B: DRV8833 H-BRIDGE -> N20 LEAD-SCREW GEARMOTOR



CURRENT SENSE: LOW-PASS FILTER + NON-INVERTING AMP (G = 11) -> RA0



POSITION FEEDBACK: SLIDE POT (OFF-BOARD) -> RC FILTER -> RA1



Kaushik Gunasekaren (KG)

Team 102: Modular Spider Leg

Sheet: /

File: Subsystem Schematic KG.kicad_sch

Title: Driver Nano 3 (Linear Actuator B + IMU) Subsystem Schematic

Size: A3	Date: 2026-10-21	**Rev: 1**
KiCad E.D.A. 10.0.7		Id: 1/1